RedTeamKit v3.2

ATT&CK v20.0 synchronization, generated profiles, and behavioral safety validation.

What changed in v3.2

v3.0 — CORE ARCHITECTURE

Standard Library Foundation

Introduced the stdlib-only engine for planning, detection validation, and reporting. Static ATT&CK seed.

v3.1 — PLATFORM REFINEMENT

Deployment & CI

Expanded reporting capabilities and integrated full CI testing across Python 3.9–3.13.

v3.2 — DATA DYNAMICS

ATT&CK Sync & Validation

Living STIX 2.1 parser, generated profiles, strict behavioral safety, and verified v20.0 synchronization.

THE VERDICT

From a static planning tool to a versioned data engine capable of tracking the evolving threat landscape.

The ATT&CK sync pipeline

01 — FETCH

Official STIX Source

Synchronizes with the official Enterprise STIX 2.1 collection. Validates v20.0 versioning and availability before ingestion.

→

02 — PARSE

Strict Incompatibility

Intentionally fails loudly if the bundle shape deviates from Enterprise norms. Maps Tactic IDs and kill-chain phases.

→

03 — MAP

Strategy Linking

Relates techniques to detection strategies. Links intrusion-sets (Groups) to their documented TTP lists for generation.

→

04 — DIGEST

Deterministic Index

Generates a SHA-256 digest of the canonical JSON payload to ensure index integrity and reproducibility across nodes.

DONE

Offline by default. Refresh when you choose.

THE CLI LIFECYCLE

```
# Validate local seed (offline)
redteamkit validate --json
# Preview remote v20.0
redteamkit sync --dry-run
# Refresh local index
redteamkit sync
# CI enforcement guard
redteamkit sync --offline
```

v3.2 preserves zero-config offline planning while enabling explicit synchronization with official threat intelligence.

INDEX COMPARISON

Compact Seed Index	OFFLINE
File Size	264 KB
Technique Coverage	149 (Curated)
Intrusion Groups	174 (Active)

Full Refreshed Index	LIVE
File Size	~1.5 MB
Technique Coverage	706 (Enterprise)
Detection Strategies	Linked (v20.0)

Curated expertise + generated coverage

HAND-MAINTAINED

Curated Registry

- Origin Detail
- Tool Chains
- Infra Patterns
- Fidelity Notes

DYNAMIC

Generated Index

- STIX Metadata
- Alias Mapping
- TTP Coverage
- Group Stats

The Authoritative Split

v3.2 introduces a dual-registry architecture. Curated profiles remain authoritative for high-fidelity emulation, while the generated index provides broad coverage for the full ATT&CK corpus.

```
// Precedence Example: APT29
```

Source	hand-maintained
TTP Logic	Curated Fidelity
Aliases	Cozy Bear, Nobelium ...

Safety is executable behavior

SAFETY CONTROL	V3.2 ENFORCEMENT MECHANISM	STATUS
Destructive Filtering	T1486/T1490/T1485 excluded by default in all campaign planners, including custom phases.	VERIFIED
Input Validation	CampaignError raised for profile/template mismatches; actionable guidance for seed vs sync states.	VERIFIED
Offline Guard	Sync tool intentionally fails non-zero in offline mode to prevent accidental network leaks.	STRICT
Evidence-Backing	Reporting engine refuses to generate scores without a verified detection-results JSON.	ENFORCED

>_ VALIDATION NOTE

RTK v3.2 transitions from "best effort" safety to structural enforcement, ensuring engagements stay within authorized behavioral bounds by default.

Verified, not vibes

16
UNIT TESTS PASSED

100%
RUFF / MYPY CLEAN

706
LIVE TECH COVERAGE

149
OFFLINE TECH SEED

PASS Strict STIX 2.1 parser validation (v20.0 bundle)

PASS Curated profile precedence & alias resolution

PASS Wheel distribution including binary seed data

RELEASE DIGEST

Version 3.2 represents the most thoroughly tested release to date, ensuring data integrity from sync to final report.

A better foundation for authorized testing.

```
$ git clone https://github.com/Panda1847/redteamkit
$ cd redteamkit && pip install -e .
$ redteamkit sync && redteamkit validate
```

github.com/Panda1847/redteamkit

RedTeamKit is for authorized work only. You are responsible for scope, safety, and legal compliance.